Introduction to Economics

Class 10 — Market failures, externalities and public goods

Pavel Kocourek

Universitat de Barcelona · Group A6

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Slide 1 Title

Block 3's third class. Class 9 looked at a market that works as it should and a government that steps in anyway; today the markets themselves get it wrong. First, deals that hurt or help people who were not part of them, and the tools that can put that right: a tax, a bargain, a rule, permits. After the break, goods nobody can be kept from using, which a market provides too little of or uses up too fast. The thread is a real price on CO_2: since 2005 every large plant in the EU must hand in a permit for each tonne, and the question is what that did.

Quiz 7

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet; a question marked + can have more than one correct letter; write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

10:00	Quiz 7
10:11	Warm-up
10:16	Since 2005: a permit for every tonne of CO_2
10:19	When a deal hits a bystander
10:22	Too much glass: the lost triangle, and a tax that fixes it
10:34	In pairs: vans in the old town
10:40	Too few flu jabs; the noisy terrace
10:50	Break
11:00	Rules, taxes or permits?
11:12	The EU's permits: what happened
11:15	In pairs: two power plants
11:21	Four kinds of goods; public goods
11:30	Fish in a bay; in pairs: lights for a courtyard
11:40	In one line; Problem set 2; before Thursday 29 October

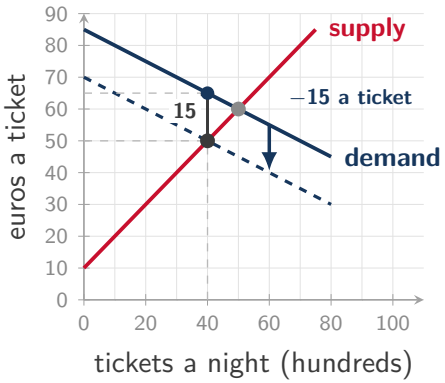
Slide 2 Plan for today

The class at a glance. Up to the break the market makes too much or too little of something because a bystander is left out of the price: a glassworks' smoke, delivery vans, flu jabs, a noisy terrace. After the break, what a government can do when the people involved cannot settle it among themselves, a rule, a tax or tradable permits, and then the goods that no market handles well at all, from Christmas lights to fish in a bay.

Two halves of fifty minutes with a ten-minute break; Problem set 2 is handed out at the end.

Warm-up: a tax paid by the buyers — who pays what?

Flamenco shows: a tax of 15 a ticket



1. Tickets now, and what venues keep?
   4,000 a night at 50: the new crossing
2. What does a buyer pay in all?
   50 + 15 = 65: on the solid demand
3. Who bears more of the 15, and why?
   venues: 10 less; buyers: 5 more
   venues cannot walk away: less elastic

Slide 3 Warm-up: a tax paid by the buyers — who pays what?

A warm-up on Class 8's tax collected from the buyers, read on a figure. When buyers hand over a tax of 15 on top of the ticket price, the most they will pay the venue falls by 15 at every quantity: that is the dashed line. Where it meets supply is the price the venues keep, 50, and the number sold, 4,000 tickets a night; it is not what a buyer pays, because the tax comes on top: $50 + 15 = 65$, straight up on the solid demand. Before the tax the price was 60, so buyers pay 5 more and the venues keep 10 less: the venues bear two thirds of the tax although the buyers hand it over. The side that cannot walk away bears more, and here that is the venues: for the same 1,000 fewer tickets their price fell 10 while the buyers' rose only 5, so between these two points their supply is less elastic than the buyers' demand (midpoint method: about 1.2 against 2.8).

Warm-up: consumer surplus, buyer by buyer

Four friends, one tandem paragliding flight each at most

The most each would pay: 150, 125, 100, 70 euros. A flight costs 90.

1. Who buys a flight?
   three: those who would pay 150, 125 and 100
2. Their consumer surplus?
   $60 + 35 + 10 = 105$ euros
3. Does the friend who would pay 70 count?
   no: does not buy at 90, so adds 0, not -20

Slide 4 Warm-up: consumer surplus, buyer by buyer

The second warm-up is Class 8's consumer surplus, counted buyer by buyer. At 90 a flight only the friends who would pay at least 90 buy, and each gains what they would pay minus the 90: 60, 35 and 10, so the consumer surplus is 105 euros. The 375 they would pay in all is not their gain, because the price comes off, and the 270 they pay goes to the seller. The friend who would pay at most 70 does not fly at 90, so adds nothing: there is no surplus, and no loss, for someone who does not buy.

Since 2005: one permit for every tonne of CO_2

a power station or a steelworks in the EU → **hands in one permit for every tonne of CO_2 it emits**

The EU limits the number of permits; firms trade them.
In 2024 one cost about 65 euros.

Did emissions fall — and by how much?

EU Emissions Trading System (ETS).

European Commission, Carbon market report COM(2025) 735, 3 December 2025.

Slide 5 Since 2005: one permit for every tonne of CO_2

The hook, from the European Commission's own report. Since 2005 every large power station and factory in the EU, since 2012 also flights within Europe and since 2024 shipping, together about two fifths of the EU's emissions (the Commission's figure for 2024), must hand in one permit for each tonne of CO_2 it emits in a year; the EU fixes how many permits exist and issues fewer each year, and firms that need more buy them from firms that need fewer. The slide asks what twenty years of that did to emissions. Keep the question open: after the break the deck shows what a cap does, and then the data. The idea behind the whole class is already here: one plant's CO_2 lands on everyone.

An externality: a deal between two that hits a third

buyer and seller: a deal both want → **a bystander: a cost or a benefit, unpaid**

Negative: a bar's terrace keeps the street awake

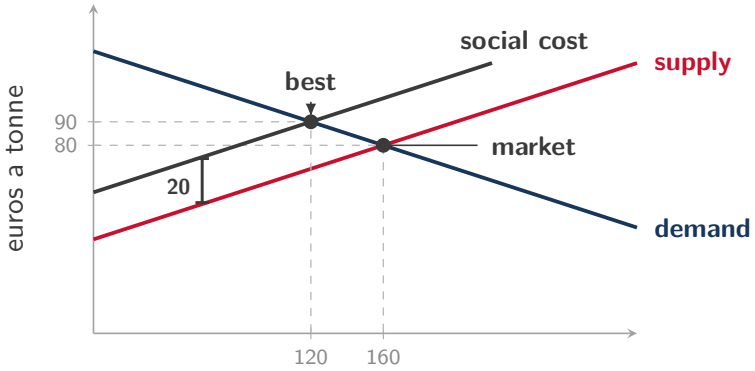
Positive: your flu jab protects your flatmates

The price counts the deal, not the bystander.

Slide 6 An externality: a deal between two that hits a third

The definition in words: an externality is the effect of a decision on a bystander who is neither paid for it nor pays for it. A harm is a negative externality, a benefit a positive one. The market price weighs only the two sides of the deal, the buyer's value and the seller's cost, so it can push the quantity too high when bystanders are hurt and too low when they gain; the next slides put numbers on both. Not every effect on others counts: a price that rises because more people want the good is felt by other buyers, but it is paid for through the market, so it is no externality.

Glass: the market makes 160; counting the smoke, 120 is best

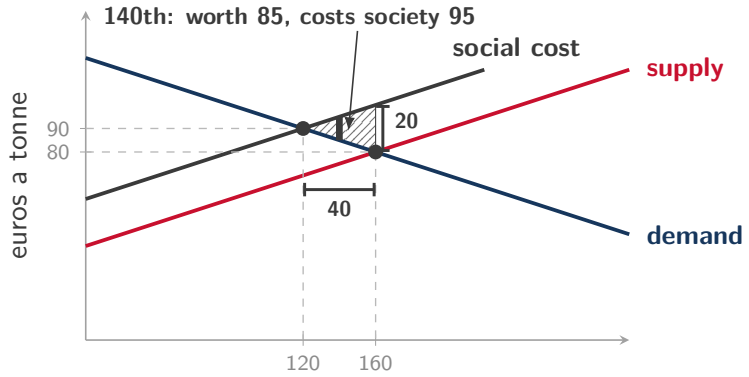


Social cost = the works' own cost + 20 euros of smoke a tonne.

Slide 7 Glass: the market makes 160; counting the smoke, 120 is best

The core picture, built in three steps. Demand is what builders and bottlers would pay for each tonne of glass; supply is the works' own cost of making it, and the market settles where the two meet. The smoke adds a cost nobody in the deal pays, so the cost to society of a tonne is the works' cost plus the harm, drawn as a line the same distance above supply everywhere. The best quantity is where the value of the last tonne just covers its full cost, and it is smaller than the market's but not zero: glass is worth having, so the best amount still makes some smoke. The social cost is not the harm alone, which is a common slip.

Tonnes 120 to 160 cost society more than they are worth: 400 thousand euros a year

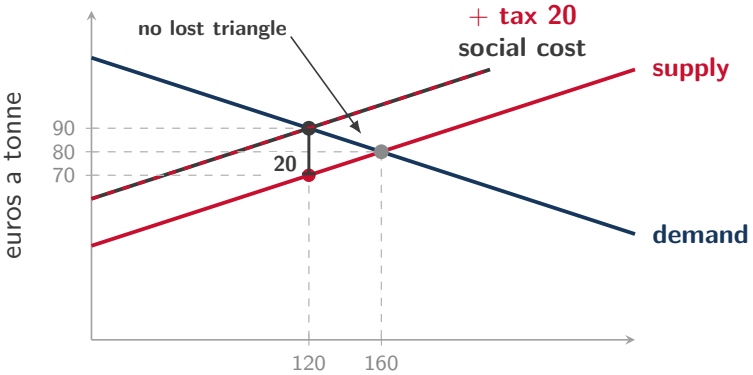


Lost: ½ × 20 × 40 = 400 thousand euros a year — Class 9's triangle, now from too much.

Slide 8 Tonnes 120 to 160 cost society more than they are worth: 400 thousand euros a year

Why the extra tonnes are a loss and not just a nuisance. Each tonne between 120 and 160 is worth less to its buyer than it costs the works and the town together, yet the market makes it, because the two in the deal only see their own gain. The losses start at nothing at 120 and grow to the full harm per tonne at 160, which is why they form a triangle, and half of base times height is the same formula as Class 9's. The difference is the direction: there a tax stopped deals worth making, here the market makes deals not worth making. The loss is not the harm on all 160 tonnes; the first 120 are worth more than their full cost.

A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone



Pigovian tax: a tax that corrects a negative externality; set at the external cost (20).

Slide 9 A tax of 20 a tonne: the market stops at 120, and the lost triangle is gone

The fix, and the reason it is exactly 20. With the tax the works pay their own cost plus the harm, so their supply curve moves up by 20 and lands on the social cost line: the deal now counts the bystander, which the reading calls internalising the externality. The market then finds 120 on its own, buyers pay 90 and the works keep 70, and the town collects the tax on every tonne still made. This is a Pigovian, or corrective, tax: a tax that corrects a negative externality, here set at the external cost per unit, not at the whole social cost, which the works already bear. In Class 9 a tax on coffee created a lost triangle because that market had nothing wrong with it; here the market was wrong to begin with, and the tax removes the loss.

In pairs, four minutes: vans in the old town

Deliveries in the old town

deliveries a day	last one is worth	last one costs the firm
160	22	14
200	20	16
240	18	18

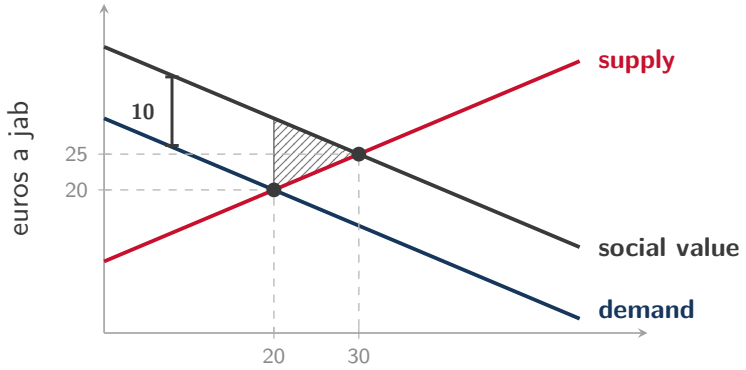
Each delivery's fumes and noise cost residents 4 euros.

1. How many deliveries does the market make?
   240: worth 18 = cost 18
2. How many are best, counting the residents?
   200: worth 20 = 16 + 4
3. Which tax gets the market there?
   4 euros a delivery
4. The lost surplus from making 240, not 200?
   ½ × 4 × 40 = 80 euros a day

Slide 10 In pairs, four minutes: vans in the old town

Exercise 1 is slides 7 to 9 on a table instead of a graph. The market keeps adding deliveries while the shops' worth covers the firm's own cost; counting the residents, each delivery costs 4 more, so the best quantity is the row where worth equals the firm's cost plus 4, and a tax of exactly that 4 makes the firm stop there on its own. The lost surplus is the triangle between the best and the market quantity: its base is the 40 deliveries too many, not 200 or 240, and its height is the harm per delivery; half of the product is the answer. The best number is not where worth equals the harm alone, and the tax is not the whole cost of a delivery.

Flu jabs protect others too: the market gives 20 thousand, society wants 30



Too few: a subsidy of 10 a jab gets the market to 30.

Slide 11 Flu jabs protect others too: the market gives 20 thousand, society wants 30

The mirror image of the glass. Your demand counts the protection you get; the jab also protects the people around you, so society's value curve sits above demand by the benefit to others, and the best quantity is where that curve meets the cost of a jab. The market stops short of it: the jabs between the two quantities are worth more to everyone than they cost and are never given, and their lost triangle is slide 8's formula again, half of 10 times the 10 thousand jabs not given. The fix mirrors the tax: a subsidy equal to the benefit to others, which is why health systems give flu jabs free to older people and risk groups. In Class 8 a subsidy created a lost triangle; here it removes one.

The noisy terrace: does it matter who holds the right?

A bar's terrace, open late

The bar gains 200 a night · the noise costs the 30 neighbours 300 a night

1. The bar has the right to stay open. What happens?
   the neighbours pay it 200 to 300 to close
2. The neighbours have the right to quiet. Now?
   it closes: the bar would offer at most 200
3. The bar gains 400 a night instead: open or closed?
   open either way: the right decides who pays

Slide 12 The noisy terrace: does it matter who holds the right?

The terrace promised in Class 9, and a way out that needs no state. The bar gains less from the late hours than the neighbours lose, so closing is the efficient outcome, and if the two sides can bargain they reach it whichever of them holds the right: when the bar may stay open, the neighbours can pay it more than it earns and less than they lose; when they hold the right to quiet, the bar cannot offer enough to buy it. What the right settles is who pays whom, not what happens. If the terrace earned the bar more than the neighbours lose, it would stay open either way, again with the right settling who compensates whom. This is the Coase theorem: with no cost to bargaining, the people involved fix the externality themselves.

A tax on plastic cups: which curve moves? (optional)

A tax of 0.10 on every plastic cup, paid by the makers

1. Which curve shifts, which way?
   supply of cups: up by 0.10
2. Cups get dearer: does demand shift too?
   no: fewer cups bought, along demand
3. Reusable cups get cheaper. Now?
   demand for plastic cups shifts left

Slide 13 A tax on plastic cups: which curve moves? (optional)

An optional exercise on Class 5's shift question, tied to slide 9's tax. A tax paid by the makers raises their cost of every cup, so the supply curve moves up by the tax, which is the same as moving left. The higher price then makes buyers take fewer cups, but that is a move along the demand curve already drawn, not a second shift: a good's own price never shifts its own curve. A cheaper reusable cup is a substitute, so fewer plastic cups are wanted at every price and demand does shift, to the left.

Break

Back at 11:00

slide 14 of 27

Slide 14 Break

Ten minutes. After the break: why thirty neighbours rarely strike the terrace bargain, and what a council can do instead.

Thirty neighbours rarely bargain: so the council sets a rule or a price

- Thirty neighbours: who organises them, who pays the lawyer?
- Each hopes the others will pay
- Each side holds out for a better deal
- So the council steps in: a closing hour, or a charge

slide 15 of 27

Slide 15 Thirty neighbours rarely bargain: so the council sets a rule or a price

Why slide 12's bargain stays on paper when the bystanders are many. Getting thirty people into one room, agreeing who pays what and paying a lawyer are transaction costs that can exceed what the deal is worth; each neighbour's quiet also benefits all the others, so each waits for someone else to pay, a problem that returns under its own name with the Christmas lights; and once both sides hold out for a better split, the deal can fall apart even though one exists. When private bargaining fails, the state can act for the many, either with a rule, such as a closing hour or an emissions limit, or with a price, such as a tax on each unit of harm. The next slides compare the two.

Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

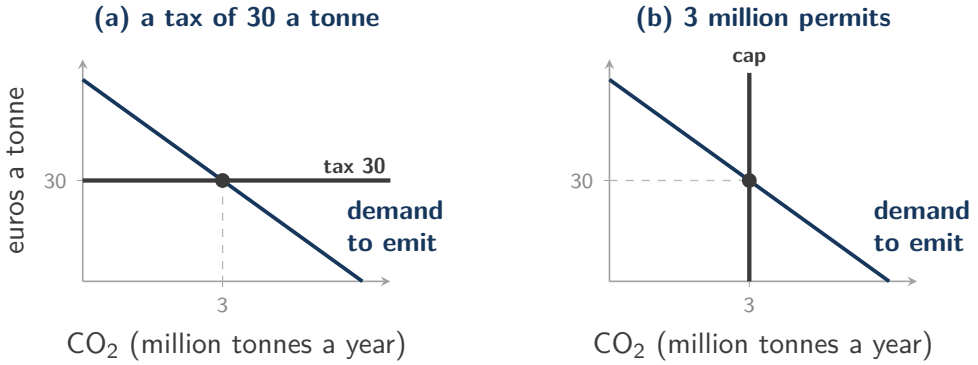
	Works A	Works B	cost of cutting (thousand euros)
CO_2 now (thousand tonnes a year)	4	4	
cutting a tonne costs (euros)	10	40	
Rule: each cuts 2	2×10	2×40	**100**
Tax 25 a tonne emitted	cuts 4	cuts none	$\mathbf{4 \times 10 = 40}$

The cheap cutter does the cutting.

Slide 16 Cutting 4 thousand tonnes costs 100 under a rule, 40 under a tax

Why a price beats a rule for the same cut. The rule makes each works cut the same amount regardless of what cutting costs it, so the works for which cutting is dearest does as much of it as the cheap one. Under the tax each works compares cutting with paying: A cuts because cutting is cheaper than the tax, B pays because paying is cheaper than cutting, and the whole cut lands on the cheap cutter. B's tax payment is not part of the cost of the cut; it goes to the town, which can lower other taxes with it. The town did not need to know which works cuts cheaply, since the tax sorts that out, though it did need a tax somewhere between the two cutting costs, and because the tax keeps charging for every tonne still emitted it also rewards a cleaner technology, where a rule stops at the limit; whether equal cuts are fairer is a different question from whether they are cheaper.

A tax fixes the price of emitting; a permit cap fixes the quantity

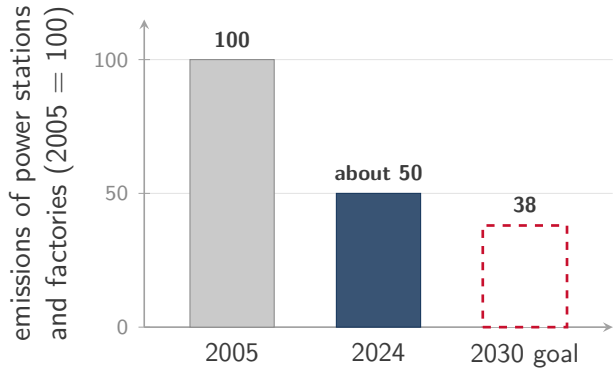


Same point: a tax picks the price, a cap picks the quantity.

Slide 17 A tax fixes the price of emitting; a permit cap fixes the quantity

The two prices on the same picture. The line is every firm's tonnes lined up from the dearest to cut to the cheapest, so its height is what a firm would pay to emit one more tonne, which is what cutting that tonne would cost it. With a tax, each firm emits the tonnes that cost more than the tax to cut and cuts the rest, so the quantity follows from the price; with permits the state fixes the quantity and trading sets the price, since firms with dear cuts buy from firms with cheap ones, and a permit ends up worth exactly what the last cut costs. Set right, the two land on the same point; they differ in what is certain, the quantity or the cost to firms. Even a firm given its permits for free pays to pollute, because it could have sold them: Class 2's opportunity cost.

EU permits since 2005: power stations and factories emit about half as much



The cap shrinks every year; the price decides who cuts.

European Commission: COM(2025) 735, 3 December 2025; verified EU ETS data for 2025, 10 April 2026.

Slide 18 EU permits since 2005: power stations and factories emit about half as much

Slide 5's answer, in the Commission's own figures for 2024, with the 2030 goal, a cut of 62 per cent from 2005, drawn as the dashed bar; in 2025 the permit system's emissions fell another 1.3 per cent. What the cap guarantees is the number of tonnes; the price then decides which firms cut, as slide 17 said. How much of the fall is the permit price itself is a harder question: the Commission credits the recent cuts to renewable and nuclear power replacing coal and gas, which happened partly because of the permit price and partly because solar and wind got cheap on their own, and careful studies disagree on the split. So the honest reading is that the cap fixes how many tonnes may be emitted, while how much of the fall so far is the permits' own doing is debated.

In pairs, four minutes: two power plants, a rule, a tax, permits

	Plant N	Plant S
CO_2 now (thousand tonnes a year)	6	6
cutting a tonne costs (euros)	20	50

Goal: 6 thousand tonnes less in all. Costs in thousand euros.

1. Rule: each cuts 3. Cost?
   $3 \times 20 + 3 \times 50 = 210$ thousand euros
2. Tax 30 a tonne emitted: who cuts?
   N cuts all 6; S cuts none
3. The cost of the cut with the tax?
   $6 \times 20 = 120$ thousand euros
4. Tradable permits to emit 3 each: who sells?
   N sells 3 to S, at 20 to 50 a tonne

Slide 19 In pairs, four minutes: two power plants, a rule, a tax, permits

Exercise 2 is slide 16 with a third policy. Under the rule each plant cuts half the goal whatever it costs, so the dear cutter does half the work; under the tax each plant compares cutting with paying, and only the plant whose cut is cheaper than the tax cuts, which here means the whole goal from one plant at well under the rule's cost. The tax the other plant pays is not part of the cost of the cut. Tradable permits reach the same place by another road: the plant with the dear cut would pay up to its own cutting cost for a permit, the plant with the cheap cut would sell for anything above its own, so permits move from the cheap cutter to the dear one at a price between the two costs, and the cut is made where it is cheapest whoever got the permits at the start. That is slide 12's Coase point again: the first allocation decides who gets the money, not who cuts.

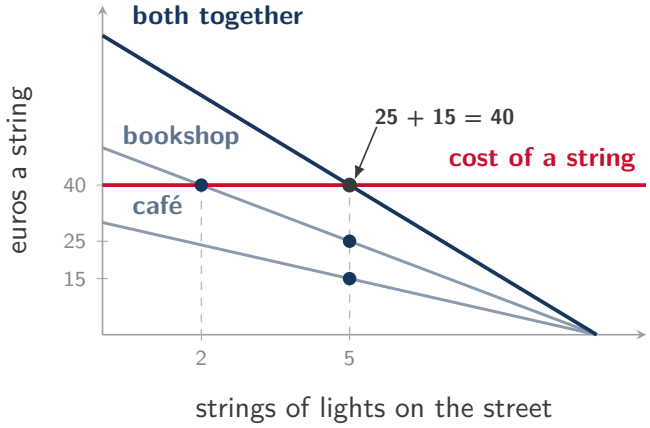
Two questions sort goods into four kinds

	excludable (non-payers kept out)	**not excludable** (anyone can use it)
rival (my use leaves less for you)	**private good** a cup of coffee	**common resource** fish in a bay
not rival (it does not)	**club good** a toll motorway, no jams	**public good** a street's Christmas lights

Slide 20 Two questions sort goods into four kinds

The other side of market failure starts with a sorting rule. Excludable means those who do not pay can be kept out; rival means my use leaves less for you. Markets handle goods that are both, because the seller can charge and each unit goes to one buyer; each of the other three boxes fails in its own way, and the next slides take the two that are open to everyone. The same good can change box: a free road is a public good at four in the morning and a common resource at rush hour. A public good is not simply anything the state provides; it is defined by the two questions, whoever supplies it.

A public good: add what each would pay for each string — the street wants 5



Both get the same lights: add what each would pay.

Slide 21 A public good: add what each would pay for each string — the street wants 5

Why the market lights too little of the street. Each shop alone hangs strings while a string is worth the cost to it, and the café never starts. But every string lights the whole street, so a string's value to the street is what the bookshop would pay plus what the café would pay for that same string, and that sum covers the cost of a string further out than either shop alone. The gap is the free rider: the café enjoys the bookshop's strings without paying and the bookshop ignores the café's value, so the strings worth hanging between 2 and 5 are not hung, and the fix is to make everyone pay a share, by a club or by taxes, as a council does with fireworks. For a private good such as a coffee we add the other way, quantities at one price across buyers, because each cup goes to one person; for a public good we add values for the same unit, because everyone gets it.

A free bay fills with boats: each new boat takes fish from the others

Boats in a bay

boats	catch in all	per boat
	(thousand euros of fish a week)	
1	12	12
2	22	11
3	30	10
4	36	9
5	40	8
6	42	7

A boat's week costs 7.

4th boat: catches 9, but the bay gains only 6

Open bay: six boats, each just covers its 7

Best: three boats; $30 - 21 = 9$ a week left

Rival, open to all: used too much.

Slide 22 A free bay fills with boats: each new boat takes fish from the others

The other open-access failure, rival this time. A skipper sees his own catch, but the bay's total catch rises by less than that, because part of what the new boat lands is fish the boats already there would have caught: that shortfall is the externality inside a common resource. Since nobody can be kept out, boats keep coming while a boat's own catch covers its cost, until the whole catch is eaten up by costs and nothing is left over. For the bay, a boat is worth adding only while it adds at least its cost to the total catch, which stops much earlier. This is the tragedy of the commons; the overuse comes from open access and rivalry, not from greed, and the fixes all restore one or the other: licences or quotas, a charge per boat, or an owner for the bay.

In pairs, three minutes: lights for a shared courtyard

Two flats share a courtyard; a string costs 25

	1st	2nd	3rd	4th string
Flat A would pay	30	20	10	0
Flat B would pay	20	15	10	5

1. Flat A alone: how many strings?
   1: the 2nd is worth 20, less than 25
2. Flat B alone?
   none: 20 is less than 25
3. Both together?
   2: the 2nd is worth 35; the 3rd only 20
4. Why might they end up with fewer?
   each counts only its own value: B free-rides

Slide 23 In pairs, three minutes: lights for a shared courtyard

Exercise 3 is slide 21 on a table. Alone, each flat hangs a string while that string is worth at least its cost to that flat by itself; together, because both see every string, the test is whether what the two would pay for the same string adds up to the cost, and that keeps going one string further. The number they end up with is lower than the best because each flat counts only its own value: the flat that would not buy alone enjoys the other's string for nothing, and the other will not pay the full cost of a second string worth less than that to it. The traps are adding strings across flats, which is the private-good sum, and adding values of different strings.

What does a morning's beach clean-up cost you? (optional)

Saturday: 4 hours cleaning the beach

You could earn 11 an hour instead · the bus to the beach costs 3
Your monthly gym fee is already paid

1. Your opportunity cost of the morning?
   $4 \times 11 + 3 = 47$ euros
2. Is the gym fee part of it?
   no: paid whatever you do (sunk)
3. Why do too few people volunteer?
   the clean beach is for everyone: a public good

Slide 24 What does a morning's beach clean-up cost you? (optional)

An optional exercise on Class 2's opportunity cost, tied to slide 21. The cost of the morning is what you give up by spending it on the beach: the wages you could have earned plus the bus fare you pay only because you go. The gym fee is paid whether you clean or stay in bed, so it is no part of the choice, like money already paid and gone in Class 2: sunk. Too few people volunteer for the same reason the shops hang too few lights: the person who cleans bears the whole cost, while the clean beach is enjoyed by everyone, including those who stayed in bed.

A tax on petrol: positive or normative? (optional)

A tax on petrol: three statements

1. "A tax of 0.20 a litre cuts driving more than one of 0.10."
   positive: data can settle it
2. "Cleaner air is worth dearer driving."
   normative: a judgement of values
3. Both want cleaner air. B: "Car-free zones instead: drivers barely react to prices."
   B's reason: values or a prediction?
   a prediction: how drivers respond

Slide 25 A tax on petrol: positive or normative? (optional)

An optional exercise on Class 4's distinction, on the taxes of slides 9, 16 and 17. A claim about what a tax would do to driving is about the world and evidence can settle it, whether or not it mentions a tax or a number; a claim that cleaner air is worth dearer driving weighs one thing against another and no data can settle it. The third item is the trap: two people who share a goal can still give opposite advice because they predict differently how much drivers react to a price, and that disagreement is positive, not a clash of values.

A cost on a bystander:
the market makes too much;
a tax equal to that cost fixes it.

Open to all: a public good gets too little,
a common resource is used too much.

A price lets the cheapest cutter do the cutting.
Thursday: what it costs a firm to make things.

slide 26 of 27

Slide 26 In one line

The class in four lines, to read slowly: the first is slides 6 to 10, the second slides 20 to 23, the third slides 16 to 19, and the fourth names Thursday.

Before Thursday 29 October

- Thursday: production and costs
- Read Mankiw & Taylor: Ch. 5, pp. 105–122
- Problem set 2 (Block 3): handed out today; due Monday 2 November, on paper, in class
- Practise in Sextant: the units the course page lists

pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ

Pavel Kocourek Introduction to Economics · Class 10 27

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Slide 27 Before Thursday 29 October

Thursday 29 October: what it costs a firm to make things, in the short run and the long run: production and costs. Reading: Mankiw & Taylor chapter 5, pages 105 to 122. Problem set 2, on Block 3, is handed out in class and is due on Monday 2 November, on paper, in class. Practise in Sextant with the units the course page lists.

pkocourek.com/teaching/intro-economics · pkocourek@ub.edu · pkocourek.com/sextant/?c=intro-econ